Listing 4.42 Searching in `search_as_you_type` and its subfields

```
GET tech_books4/_search
{
  "query": {
    "multi_match": {
      "query": "in",
      "type": "bool_prefix",
      "fields": ["title","title._2gram","title._3gram"]
    }
  }
}
```

As the query shows, we are creating a `multi_match` query (with a `bool_prefix` as the type) because the search is carried out across the main field and its subfields (`title` and subfields of `title`: `_2gram`, `_3gram`, etc.). This query should return the books *Elasticsearch in Action* and *Elastic Stack in Action*. We use a `multi-match` query because we are searching for a value across multiple fields: `title`, `title._2gram`, and `title._3gram`.

N-grams, edge n-grams, and shingles

If you are hearing about n-grams, edge n-grams, and shingles for the first time, these concepts may baffle you. I briefly explain them here, and we cover them in detail in chapter 7.

An *n-gram* is a sequence of words of a given size. We can have 2-grams, 3-grams, and so on. For example, if the word is *action*, the 3-grams (n-grams for size 3) are `["act", "cti","tio","ion"]`, the bi-grams (size 2) are `["ac", "ct","ti","io", "on"]`, and so on.

Edge n-grams are n-grams of every token, where the start of the n-gram is anchored to the beginning of the word. For *action*, the edge n-grams are `["a", "ac","act", "acti","actio","action"]`.

Shingles are word n-grams. For example, the sentence "Elasticsearch in Action" outputs the shingles `["Elasticsearch", "Elasticsearch in", "Elasticsearch in Action", "in", "in Action", "Action"]`.

Sometimes we may want a field that we can declare as more than a single data type. For example, a movie title can be both `text` and `completion` data types. Fortunately, Elasticsearch allows us to declare a single field with multiple data types. Let's find out how we can create multi-typed fields.

4.7 Fields with multiple data types

We've seen that each field in a document is associated with a data type. However, Elasticsearch is flexible and also lets us define fields with multiple data types. For example, the `subject` field in our email data can be a `text`, `keyword`, or `completion` type,